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Major Project (EI481P)

Semester End Examination

Title: Design and Implementation of an FPGA-Based Seizure Detection System

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INTRODUCTION

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What is Epilepsy?

A neurological disorder characterized by recurrent, unprovoked seizures caused by abnormal electrical discharges in the brain.

Diagnosis: Electroencephalography (EEG) is the gold standard for monitoring brainwaves.

The Shift to Edge Computing:

- * Traditional detection uses bulky computers.
- * Wearable health devices require real-time monitoring directly on the patient (Edge Computing).
- * FPGAs (Field Programmable Gate Arrays) offer the perfect platform due to their massive parallel processing and ultra-low power consumption.



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PROBLEM STATEMENT

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Current machine-learning seizure detection systems rely on standard CPUs/Microprocessors.

The Issues:

- * **High Latency:** Operating systems introduce processing delays.
- * **High Power Consumption:** Complex algorithms drain batteries quickly, making them unsuitable for 24/7 wearable medical patches.

The Need:

A purely digital, hardware-based approach that detects seizures instantly while consuming a fraction of a Watt.



LITERATURE REVIEW

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Author(s)	Year	Title	Description
Valentino et al.	2025	Analysis of EEG records in an epileptic patient using wavelet transform	Achieved high accuracy but required significant hardware resources (DSPs).
Sayeed & Kumar	2023	Automated EEG analysis of epilepsy: A review	Addressed power consumption by removing redundant logic
Shoka, A., et al.	2023	An Efficient FPGA-Based Approach for Automatic Epileptic Seizure Detection Using EEG Signals	Presented an FPGA implementation using Hjorth descriptors and KNN classifier on Spartan-6, achieving 90.74% accuracy with fast classification and low resource utilization (<10% LUTs).
Hossain, K.M., et al.	2023	FPGA-Based Multi-Channel EEG Seizure Detection System	Developed a multi-channel EEG seizure detection system using DWT and SVM on Zynq-7000 FPGA, achieving 88.6% accuracy with efficient processing and moderate power consumption.



LITERATURE REVIEW

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Author(s)	Year	Title	Description
Chen, Z., et al.	2024	Wearable Epilepsy Seizure Detection Using Spiking Neural Networks on FPGA	Proposed an ultra-low-power FPGA implementation using spiking neural networks (SNN), achieving very fast inference time (0.5 μ s) and extremely low energy consumption, suitable for wearable systems.
Daoud, H. & Bhatt, M.A.	2023	Automated Seizure Detection on FPGA Using Artix-7	Demonstrated a threshold-based seizure detection system on Artix-7 FPGA with high accuracy (~98%) and very low power consumption, though with relatively higher detection latency.
Rafiee, J., et al.	2011	Wavelet basis functions in biomedical signal processing	Identified Daubechies-4 (db4) as an optimal wavelet for EEG signal analysis due to its superior time-frequency resolution for detecting epileptic patterns.
Park, J., et al.	2023	Resource-Optimized DWT on Virtex-7 FPGA	Proposed an optimized DWT architecture using polyphase structure, reducing area and power while improving processing speed for FPGA-based implementations.



IDENTIFIED GAPS IN EXISTING SYSTEMS:

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1. **Power-Accuracy Trade-off:** Systems achieving high accuracy often consume more power, while ultra-low-power systems sacrifice detection accuracy.
2. **Limited Multi-channel Support:** Most implementations process single-channel EEG, limiting clinical utility.
3. **High Detection Latency:** Several systems exhibit latencies >3 seconds, which is too slow for responsive neurostimulation.
4. **Lack of Adaptability:** Fixed-algorithm implementations cannot adapt to patient-specific seizure patterns.

Our Project's Position: We bridge the gap by completely eliminating the microprocessor, using an $O(1)$ algorithm that requires less than 1% logic utilization, while still providing a modern GUI for doctors.



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OBJECTIVES

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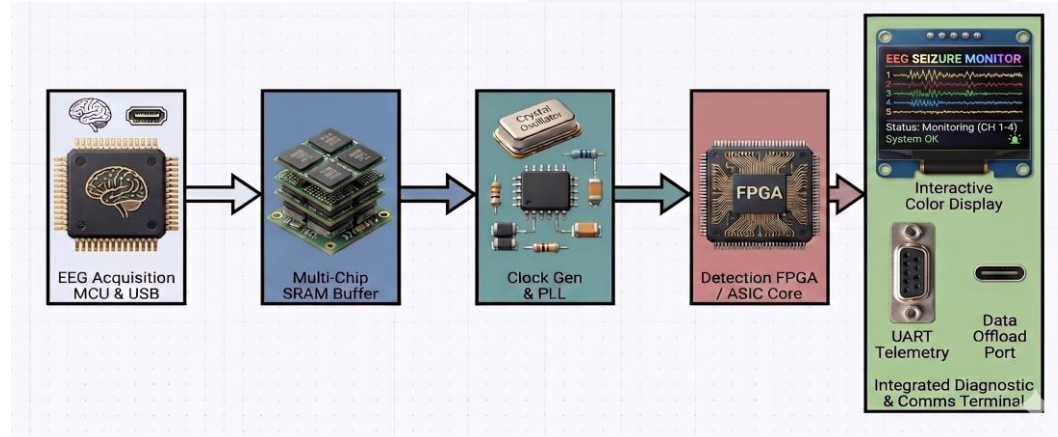
1. To develop a bare-metal, purely digital seizure detection circuit in Verilog HDL.
2. To implement a detection algorithm utilizing less than 1% of the FPGA's logic resources.
3. To build a custom UART protocol to stream processed medical data to a PC in real-time.
4. To create a Python PyQtGraph dashboard that actively monitors the hardware and triggers critical alerts instantly.

PROPOSED METHODOLOGY

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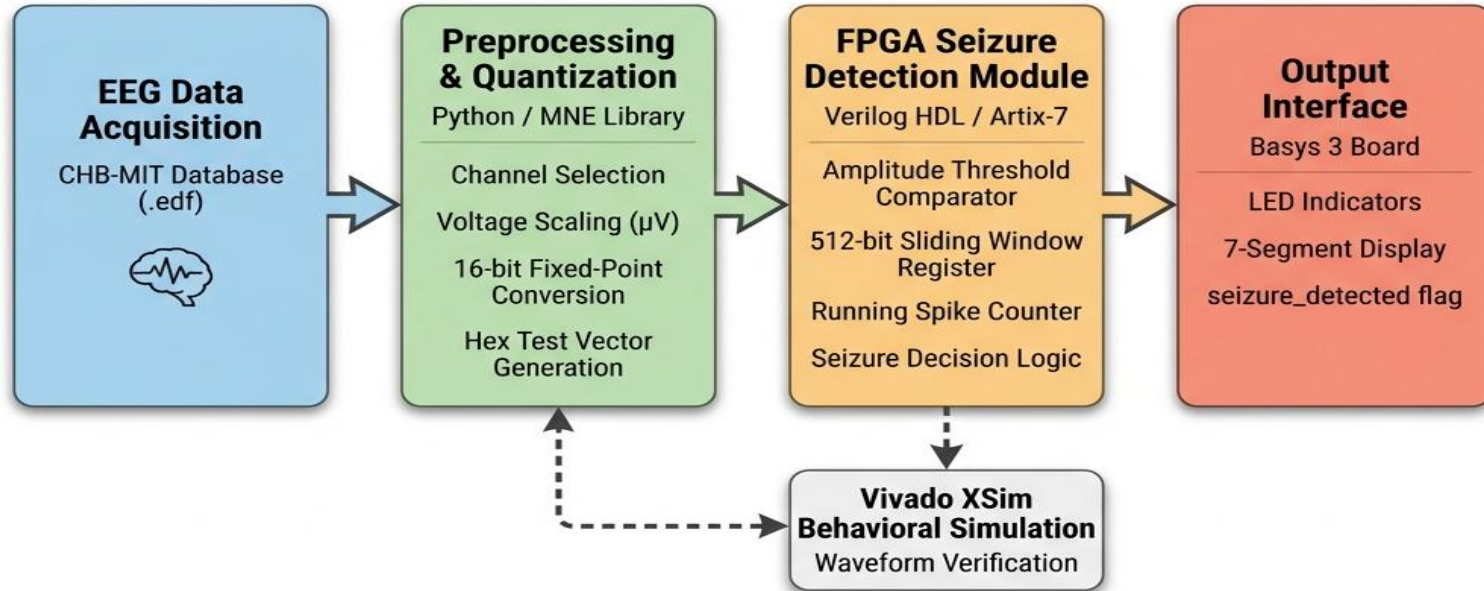
Hardware-Software Co-Design Pipeline:

1. **Data Acquisition:** CHB-MIT Scalp EEG Dataset (16-bit, 256 Hz) loaded into FPGA Block RAM.
2. **Hardware Detection:** Sliding Window Spike Counter executes in 10 nanoseconds.
3. **Local Display:** Multiplexed 7-Segment Display counts live spikes on the board.
4. **Transmission:** UART Protocol serializes data at 115200 baud.
5. **Software Dashboard:** Python reads the COM port and plots the live EEG wave and Seizure Flag.





SYSTEM ARCHITECTURE





THE CORE ALGORITHM

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The Sliding Window Spike Counter

- * **Amplitude Thresholding:** Normal brainwaves = $\pm 100 \mu\text{V}$. Seizure spikes = $> \pm 200 \mu\text{V}$.
- * **Temporal Window:** Analyzes 512 continuous samples (exactly 2 seconds of brain activity).
- * **Accumulative Thresholding:** If the window accumulates **25 spikes**, a seizure is declared.
- * **O(1) Complexity:** The hardware only performs one addition and one subtraction per clock cycle, making it insanely fast and resource-efficient.



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IMPLEMENTATION DETAILS

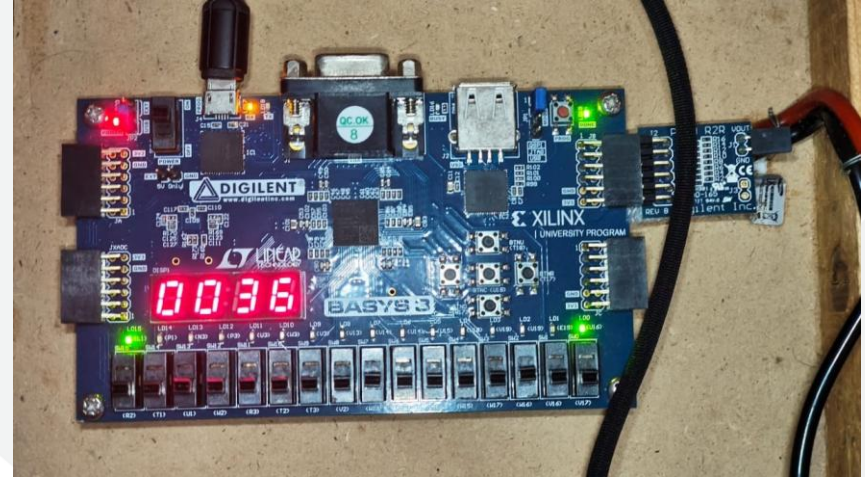
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Hardware Platform: Digilent Basys 3 Board (Xilinx Artix-7 FPGA).

Language: Verilog HDL (Synthesized using Vivado 2025.2).

Custom UART Packet:

- * 4-Byte structure ensures perfect synchronization.
- * Byte 1: Sync Byte (`0xA5`)
- * Byte 2 & 3: 16-bit EEG Data
- * Byte 4: Seizure Flag (0 = Normal, 1 = ALARM)

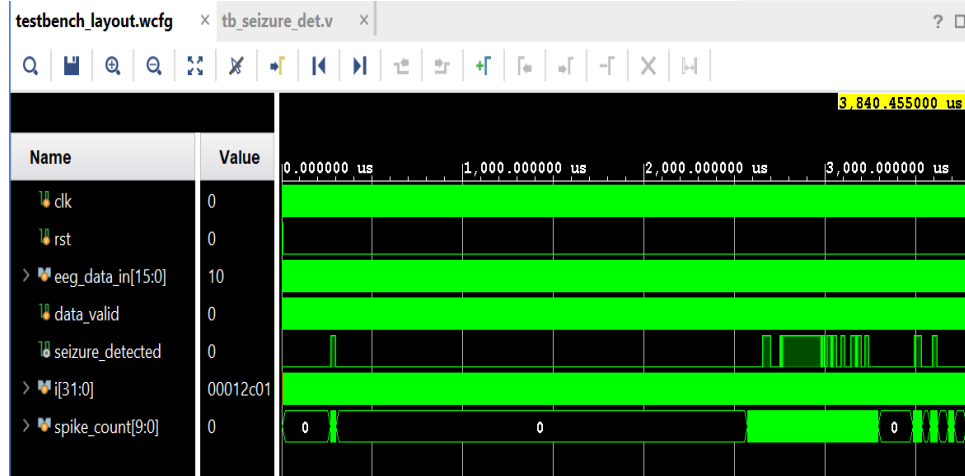




RESULTS - FUNCTIONAL VALIDATION

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Behavioral Simulation Results — Vivado XSim



* **Validation Dataset:** Tested using Patient 1 data from the CHB-MIT Database (containing a documented 40-second seizure).

* Results:

* **100% Accuracy** on the target vector.

* **0 False Positives** during the 196 seconds of normal brain activity.

* Early detection triggered within just 2 seconds of the seizure's physiological onset.



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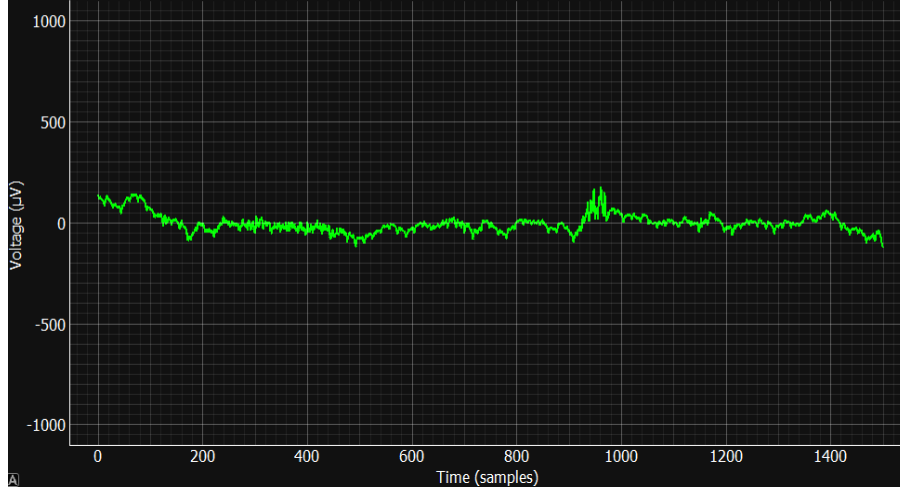
RESULTS - LIVE PYTHON DASHBOARD

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Real-Time Epileptic Seizure Detection

PATIENT STATUS: NORMAL

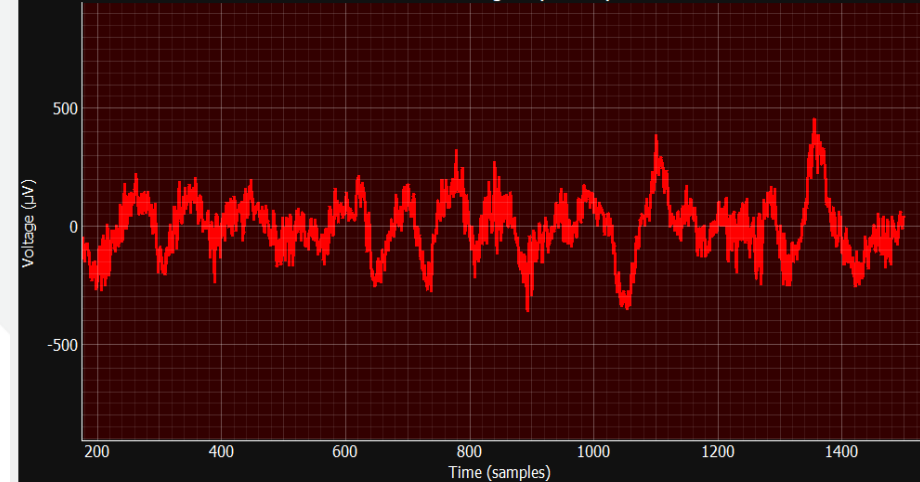
Live EEG Signal (FP1-F7)



Real-Time Epileptic Seizure Detection

⚠ SEIZURE DETECTED ⚠

Live EEG Signal (FP1-F7)

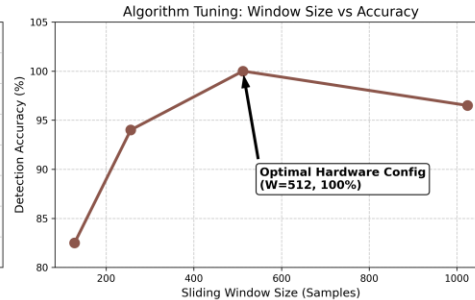
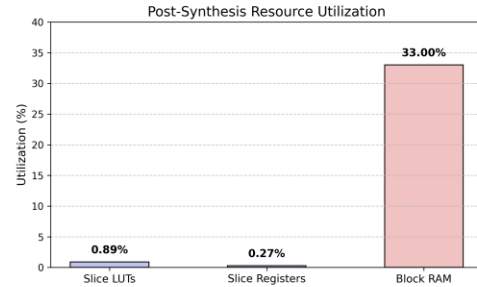
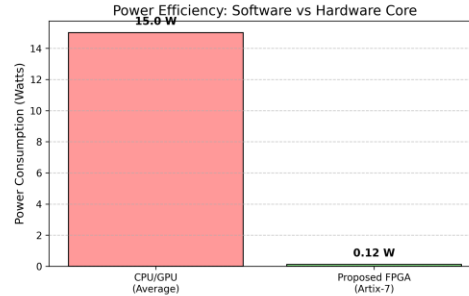
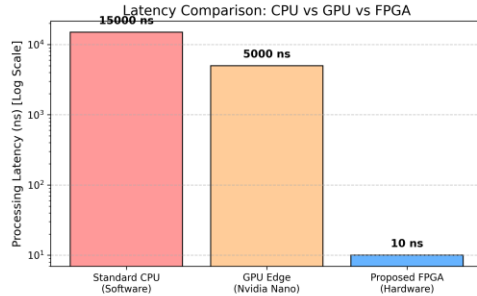




SYSTEM PERFORMANCE & RESOURCE ANALYSIS

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FPGA Seizure Detection System: Performance & Resource Analysis



The Proof of Engineering:

- * **Latency:** 10 nanoseconds (1 clock cycle).
- * **Power:** 0.12 Watts (Drastically lower than CPUs).
- * **Logic Utilization:** < 1% of LUTs and Flip-Flops used. 99% of the chip is free for future upgrades.



Conclusion:

- * Successfully developed a foundational Proof-of-Concept for wearable medical devices.
- * Proved that digital logic alone (without a CPU) can accurately identify seizures in real-time with negligible power draw.

Future Scope:

- * Scaling the logic to support multi-channel EEG arrays (Parallel processing).
- * Integrating a Bluetooth Low Energy (BLE) module to replace the wired UART connection for truly wireless ambulatory monitoring.

Learning Outcome:

1. CO1: Apply knowledge of mathematics, digital logic, and biomedical science to solve the engineering challenge of real-time epileptic seizure detection using a sliding window algorithm.
2. CO2: Design, develop, and document a multidisciplinary hardware–software system integrating FPGA-based Verilog modules, UART communication, and a live Python PyQtGraph dashboard.



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THANK YOU